Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\} + \delta x_D[k],$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\} - \delta x_D[k],$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, $\delta x_D[k]$ is the motion error caused by the disturbance, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d-step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \le \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1-\lambda_c)\,\boldsymbol{\delta x[k]} - \delta x_D[k]\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} - \delta x_D[k] = \lambda_c \cdot \delta x[k] - a_x[k] f_T[k]$$

d-step measurement delay:

$$\boldsymbol{\delta x[k]} = \delta x[k-d] + \sum_{i=1}^{d} \Delta x_d[k-i] - \sum_{i=1}^{d} a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\} - \sum_{i=1}^{d} \delta x_D[k-i]$$

Ideal control effort (with d-step measurement delay):

$$\bar{\bar{f}}_{dx}[k] = a_x^{-1}[k]\Big\{ \underbrace{\Big[\Delta x_d[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \Delta x_d[k-i]\Big]}_{\Delta x_d^d[k]} + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big]$$

$$- \underbrace{\Big[\delta x_D[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \delta x_D[k-i]\Big]}_{\delta x_D^d[k]} \Big\}$$

Motion error:

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - \underbrace{\Big\{a_x[k] f_T[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} a_x[k-i] f_T[k-i]\Big\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k]\big[f_{dx}[k] - \bar{\bar{f}}_{dx}[k]\big] - \delta x_T^d[k]$$

$$a_x[k]\,\bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^d a_x[k-i] f_{dx}[k-i]\Big] - \delta x_D^d[k]$$

$$a_x[k]\,f_{dx}[k] = \big(1 + e_{a_x}[k]\,\hat{a}_x^{-1}[k]\big)\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\big(\delta x[k-d] + n_x[k]\big) - \textstyle\sum_{i=1}^d \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}$$

$$\begin{aligned} a_x[k]\,f_{dx}[k] - a_x[k]\,\bar{\bar{f}}_{dx}[k] = {} & e_{a_x}[k]\,f_{dx}[k] + (1-\lambda_c) n_x[k] \\ & + (1-\lambda_c) \textstyle\sum_{i=1}^d \big(a_x[k-i] - \hat{a}_x[k-i]\big) f_{dx}[k-i] + \delta x_D^d[k] \end{aligned}$$

Parameter model

$$a_x[k] = a_{x_{det}}[k] + a_{x_{ram}}[k]; \qquad a_{x_{ram}}[k] \approx a_x'[k]\, x_{ram}[k]$$

$$a_x[k+1] = a_x[k] + \delta a_x[k] + \delta a_{ram}[k]$$

$$\delta a_x[k] = a_{x_{det}}[k+1] - a_{x_{det}}[k] = a_x'[k]\,\Delta h_d[k], \qquad \delta a_{ram}[k] = a_x'[k]\,\Delta x_{ram}[k]$$

$\Delta h_d[k] = h_d[k+1] - h_d[k]$ (desired-trajectory height increment).

Parameter estimator: $\hat{a}_x[k+1] = \hat{a}_x[k] + \delta a_x[k]$

Parameter estimation error: $e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$

Estimation error dynamics: $e_{a_x}[k+1] = e_{a_x}[k] + \delta a_{ram}[k]$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - \sum_{j=1}^{i} \delta a_{ram}[k-j]$$

$$\begin{aligned} a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{f}}_{dx}[k] \approx e_{a_x}[k]\, & \underbrace{\left\{ f_{dx}[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} f_{dx}[k-i] \right\}}_{F_{dx}[k]} \\ & - \underbrace{\left\{ (1-\lambda_c) \textstyle\sum_{i=1}^{d} (d+1-i)\, f_{dx}[k-i]\, \delta a_{ram}[k-i] \right\}}_{\delta x^{d}_{\delta af}[k]} \\ & + \delta x^{d}_{D}[k] + (1-\lambda_c) n_x[k] \end{aligned}$$

$$\boxed{\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] - \delta x^{d}_{D}[k] - \underbrace{\left\{ \delta x^{d}_{T}[k] - \delta x^{d}_{\delta af}[k] + (1-\lambda_c) n_x[k] \right\}}_{\varepsilon[k]}}$$

<u>State equation</u>

($\delta x^{d}_{D} \equiv 0$: disturbance not modelled)

$$\begin{aligned} \delta x_1[k+1] &= \delta x_2[k] \\ \delta x_2[k+1] &= \delta x_3[k] \\ \delta x_3[k+1] &= \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] - \varepsilon[k] \\ a_x[k+1] &= a_x[k] + \delta a_x[k] + \delta a_{ram}[k] \end{aligned}$$

<u>Output equation</u>

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} = a_x[k] - \sum_{i=1}^{d} \delta a_x[k-i] + \underbrace{\left\{ n_a[k] - \sum_{i=1}^{d} \delta a_{ram}[k-i] \right\}}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \mathbf{c}[k] = \begin{bmatrix} 0 \\ \sum_{i=1}^{d} \delta a_x[k-i] \end{bmatrix} \text{ (known)}, \qquad \mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] - \mathbf{c}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta\hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta\hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$\begin{aligned} e_{y_2}[k] &= y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \Big(\hat{a}_x[k] - \textstyle\sum_{i=1}^{d} \delta a_x[k-i]\Big) \\ &= \Big(a_x[k] - \textstyle\sum_{i=1}^{d} \delta a_x[k-i] + r_2[k]\Big) - \hat{a}_x[k] + \textstyle\sum_{i=1}^{d} \delta a_x[k-i] \\ &= e_{a_x}[k] + \underbrace{\Big\{n_a[k] - \textstyle\sum_{i=1}^{d} \delta a_{ram}[k-i]\Big\}}_{r_2[k]} \end{aligned}$$

$$\begin{aligned} \delta\hat{x}_1[k+1] &= \delta\hat{x}_2[k] + \ell_{11}[k]\, e_{y_1}[k] + \ell_{12}[k]\, e_{y_2}[k] \\ \delta\hat{x}_2[k+1] &= \delta\hat{x}_3[k] + \ell_{21}[k]\, e_{y_1}[k] + \ell_{22}[k]\, e_{y_2}[k] \\ \delta\hat{x}_3[k+1] &= \lambda_c \cdot \delta\hat{x}_3[k] + \ell_{31}[k]\, e_{y_1}[k] + \ell_{32}[k]\, e_{y_2}[k] \\ \hat{a}_x[k+1] &= \hat{a}_x[k] + \delta a_x[k] + \ell_{41}[k]\, e_{y_1}[k] + \ell_{42}[k]\, e_{y_2}[k] \end{aligned}$$

Error dynamics

$$\begin{aligned} e_{\delta x_1}[k+1] &= e_{\delta x_2}[k] - \ell_{11}[k]\, e_{\delta x_1}[k] - \ell_{12}[k]\, e_{a_x}[k] \\ e_{\delta x_2}[k+1] &= e_{\delta x_3}[k] - \ell_{21}[k]\, e_{\delta x_1}[k] - \ell_{22}[k]\, e_{a_x}[k] \\ e_{\delta x_3}[k+1] &= \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] - \ell_{31}[k]\, e_{\delta x_1}[k] - \ell_{32}[k]\, e_{a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]} \\ e_{a_x}[k+1] &= e_{a_x}[k] - \ell_{41}[k]\, e_{\delta x_1}[k] - \ell_{42}[k]\, e_{a_x}[k] + \underbrace{\delta a_{ram}[k]}_{q_4[k]} \end{aligned}$$

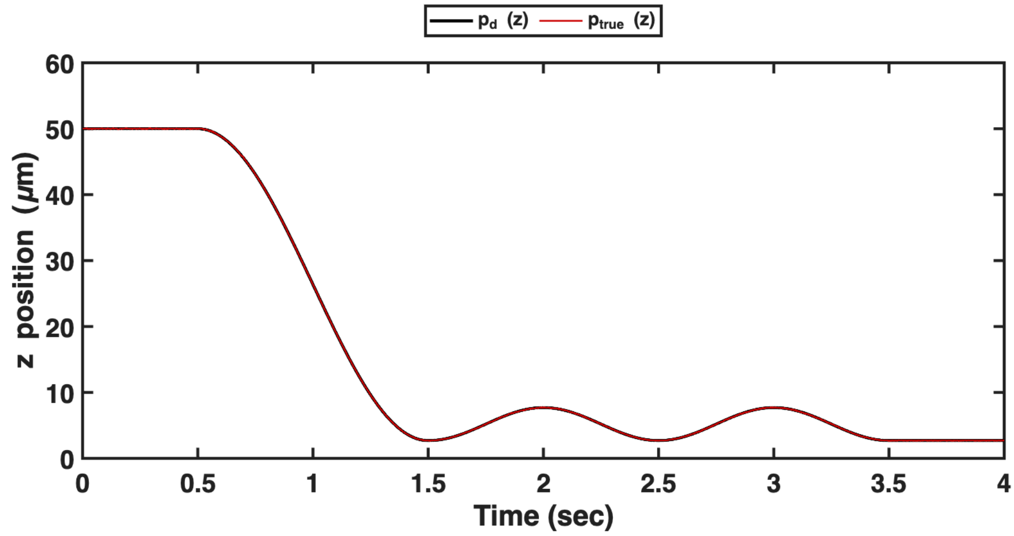
$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \lambda_c & -F_{dx}[k] \\ 0 & 0 & 0 & 1 \end{bmatrix}}$$

$$F_{dx}[k] = f_{dx}[k] + (1 - \lambda_c) \sum_{i=1}^{d} f_{dx}[k - i]$$

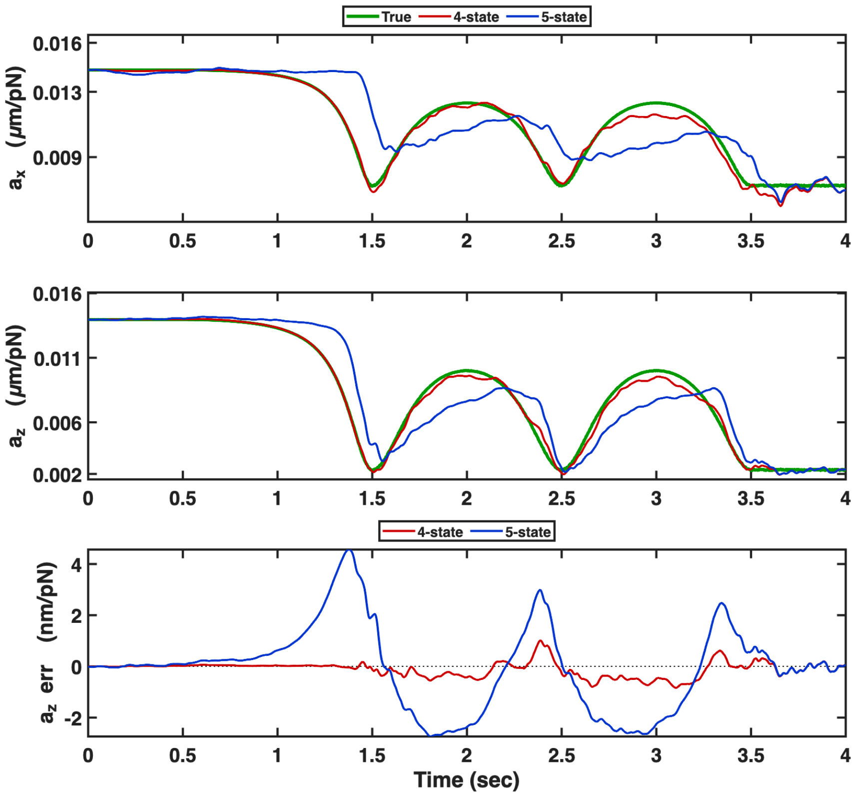
$$\mathbf{e}[k+1] = \mathbf{F}_e[k]\,\mathbf{e}[k] - \mathbf{L}[k]\,\mathbf{H}\,\mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}[k]\,\mathbf{r}[k]$$

Verification (1 Hz, near wall)

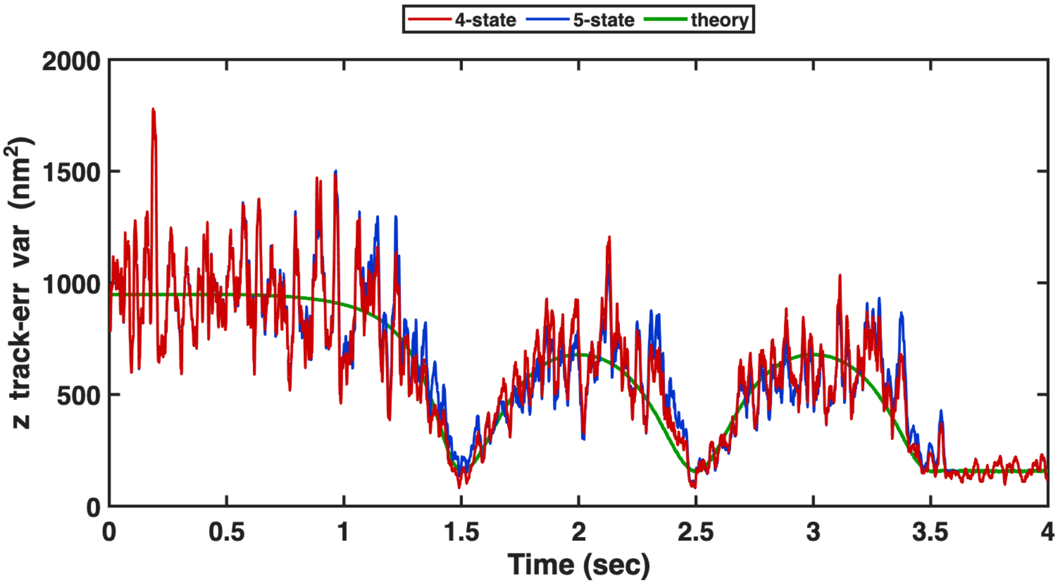
4-state vs 5-state, per-axis EKF, seed ensemble; near-wall 1 Hz oscillation ($h_{init} = 50\,\mu$m, trough $\bar{h} = 1.2$).



Desired p_d and actual p_{true} (z).



a_x, a_z: true gain and the 4-state / 5-state estimates; bottom tile, the a_z estimation error $\hat{a}_x - a_x$.



z tracking-error variance (across seeds), 4-state / 5-state, and the closed-loop theory $C_{\delta x}\,4k_BT\,a_z$ $(C_{\delta x} = 2 + 1/(1-\lambda_c^2))$.